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FIT3155: Advanced Algorithms and Data Structures

Week 8: **Amortized analysis**

Faculty of Information Technology, Monash University

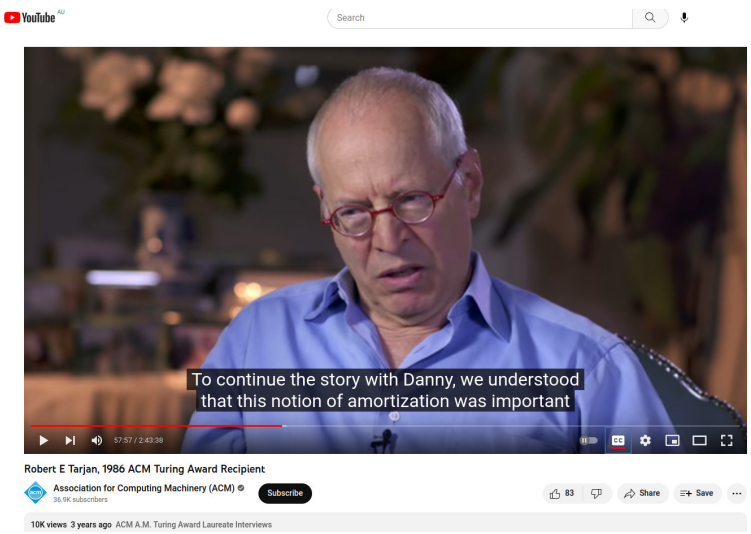
What is covered in this lecture?

Amortized analysis techniques

- Aggregate analysis
- Accounting method
- Potential method

Source material and recommended reading

- Cormen et al. "***Introduction to Algorithms***" (Chapter 17)
- Robert E. Tarjan "***Amortized Computational Complexity***" SIAM Journal on Algebraic and Discrete Methods, 6(2), 1985



Amortize?

https://www.merriam-webster.com/dictionary/amortize

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amortize verb

am·or·tize ȧ-mər-ˌtīz also ə-mȯr-

amortized; amortizing

transitive verb

1 : to pay off (an obligation, such as a mortgage) gradually usually by periodic payments of principal and interest or by payments to a sinking fund

amortize a loan

2 : to gradually reduce or write off the cost or value of (something, such as an asset)

amortize goodwill

amortize machinery

amortizable ȧ-mər-ˌtī-zə-bəl also ə-mȯr- adjective

Quoting Bob Tarjan, this term is adapted for **computational complexity** to mean "*to average over time or, more precisely, to average the running times of operations in a sequence over that sequence.*"

Motivation for Amortization in computational complexity

- In a routine use of any **data structure**, a **sequence** of operations are performed on it.
- It is useful to find bounds on the **true** time/cost to perform a **sequence** of operations.

Motivation for Amortization in computational complexity

- Summing up the **worst-case** per-operation times over a sequence of operations gives an **unduly pessimistic** estimate.
- Summing up the **best-case** per-op times over a sequence of ops yields an **unreasonably optimistic** estimate.
- Summing up **average-case** times of operations remains **infeasible** and/or **inaccurate**.
  - characterizing probability distributions over sequences of operations is non-trivial and, even if one came up with one, can lead to unrealistic estimates.

What is Amortized analysis?

An **amortized analysis** is instead used to provide a more **realistic** way to reason about the running times per operation by analyzing the bounds on the time/cost of performing any sequence of operations. (We will see how in the upcoming slides)

- Amortized analysis is about the performance of an **operation**, but averaged over any sequence of operations
- A more realistic **per-operation** time/cost can be derived, and sometimes shown to be smaller than the operation's **worst-case** behaviour.
- This is achieved by **bounding** the **true** time/cost of any possible **sequence** of operations.

Amortized analysis: three commonly used techniques

1. **Aggregate method** – a **mathematician's** view of amortization.
2. **Accounting method** – a **banker's** view of amortization.
3. **Potential method** – a **physicist's** view of amortization.

We will use a simple data structure to illustrate each of these methods.

Understanding the three methods of amortization using a simple data structure: k-bit binary counter

Counter value	A[7]	A[6]	A[5]	A[4]	A[3]	A[2]	A[1]	A[0]	Total cost
0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	1	1
2	0	0	0	0	0	0	1	0	3
3	0	0	0	0	0	0	1	1	4
4	0	0	0	0	0	1	0	0	7
5	0	0	0	0	0	1	0	1	8
6	0	0	0	0	0	1	1	0	10
7	0	0	0	0	0	1	1	1	11
8	0	0	0	0	1	0	0	0	15
9	0	0	0	0	1	0	0	1	16
10	0	0	0	0	1	0	1	0	18
11	0	0	0	0	1	0	1	1	19
12	0	0	0	0	1	1	0	0	22
13	0	0	0	0	1	1	0	1	23
14	0	0	0	0	1	1	1	0	25
15	0	0	0	0	1	1	1	1	26
16	0	0	0	1	0	0	0	0	31

A k-bit binary counter data structure

- The data structure is an boolean array of size k to store any number x in a **binary** (base-2) form.
- Let $A[0 \ldots k-1]$ be the array data structure, storing bits from the lowest ($A[0]$) to the highest ($A[k-1]$) bit order.
- The data structure is initialized to $x = 0$ in its initial state.
- This data structure supports a single operation, **increment**. This adds 1 (modulo 2^k) to the current value in the counter – see pseudocode below

The increment operation

```
b = 0
while b < k and A[b] == 1:
    A[b] = 0 /*unset*/
    b = b+1
if b < k:
    A[b] = 1 /*set*/
```

The logic of the **increment** operation

Counter value	A[7]	A[6]	A[5]	A[4]	A[3]	A[2]	A[1]	A[0]	Total cost
0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	1	1
2	0	0	0	0	0	0	1	0	3
3	0	0	0	0	0	0	1	1	4
4	0	0	0	0	0	1	0	0	7
5	0	0	0	0	0	1	0	1	8
6	0	0	0	0	0	1	1	0	10
7	0	0	0	0	0	1	1	1	11
8	0	0	0	0	1	0	0	0	15
9	0	0	0	0	1	0	0	1	16
10	0	0	0	0	1	0	1	0	18
11	0	0	0	0	1	0	1	1	19
12	0	0	0	0	1	1	0	0	22
13	0	0	0	0	1	1	0	1	23
14	0	0	0	0	1	1	1	0	25
15	0	0	0	0	1	1	1	1	26
16	0	0	0	1	0	0	0	0	31

```
1 b = 0
2 while b < k and A[b] == 1:
3     A[b] = 0 /*unset*/
4     b = b+1
5 if b < k:
6     A[b] = 1 /*set*/
```

- In each iteration of the while loop, the logic involves exploring if the bit of order b is set, i.e. $A[b] \stackrel{?}{=} 1$.
- If that bit is set (i.e. $A[b] = 1$)
  - flip/unset that bit to 0 in the array (line 3), and
  - carry forward 1 (to the next order bit b+1, handled in the next iteration of the while loop).

Worst-case analysis of increment operation

Counter value	A[7]	A[6]	A[5]	A[4]	A[3]	A[2]	A[1]	A[0]	Total cost
0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	1	1
2	0	0	0	0	0	0	1	0	3
3	0	0	0	0	0	0	1	1	4
4	0	0	0	0	0	1	0	0	7
5	0	0	0	0	0	1	0	1	8
6	0	0	0	0	0	1	1	0	10
7	0	0	0	0	0	1	1	1	11
8	0	0	0	0	1	0	0	0	15
9	0	0	0	0	1	0	0	1	16
10	0	0	0	0	1	0	1	0	18
11	0	0	0	0	1	0	1	1	19
12	0	0	0	0	1	1	0	0	22
13	0	0	0	0	1	1	0	1	23
14	0	0	0	0	1	1	1	0	25
15	0	0	0	0	1	1	1	1	26
16	0	0	0	1	0	0	0	0	31

- The **true cost** of each **increment** operation is **linear** in the **runlength** of set (1) bits before the first unset (0) bit is observed. (e.g. the counter values $\{5, 6, 7, 13, 15\}$ have **runlengths** $\{1, 0, 3, 1, 4\}$, resp.)
  - This **runlength** dictates the number of times the while loop executes, and hence its runtime (i.e. true operation cost).
- Clearly, the worst-case **runlength** is k, so the worst-case time for that operation is $O(k)$.
- Thus, if one used the worst-case to analyse the cost of a sequence of n increments one would arrive at the pessimistic bound of $O(n \times k)$ time.

Now let's analyze the amortized cost using the three methods of amortization, after introducing each.

Method 1: Aggregate method of amortization

Aggregate method: Introduction

- This method can be used when the **true time/cost** over a **sequence** of operations can be more directly **bounded** in a mathematical way, which can then be **averaged out** over that sequence length.
- In other words, in an **aggregate analysis**, one usually attempts to characterize directly a **bound** $T(n)$ on the total cost/time of a sequence of n operations of a data structure.
- If one reasons the bound for any possible sequence, the average cost per operation is estimated as $\frac{T(n)}{n}$.

Aggregate method of amortization on binary counter

Assume a sequence of n increments.

Counter value	A[7]	A[6]	A[5]	A[4]	A[3]	A[2]	A[1]	A[0]	Total cost
0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	1	1
2	0	0	0	0	0	0	1	0	3
3	0	0	0	0	0	0	1	1	4
4	0	0	0	0	0	1	0	0	7
5	0	0	0	0	0	1	0	1	8
6	0	0	0	0	0	1	1	0	10
7	0	0	0	0	0	1	1	1	11
8	0	0	0	0	1	0	0	0	15
9	0	0	0	0	1	0	0	1	16
10	0	0	0	0	1	0	1	0	18
11	0	0	0	0	1	0	1	1	19
12	0	0	0	0	1	1	0	0	22
13	0	0	0	0	1	1	0	1	23
14	0	0	0	0	1	1	1	0	25
15	0	0	0	0	1	1	1	1	26
16	0	0	0	1	0	0	0	0	31

- There are $\sim \frac{n}{2}$ counter values with $A[0] = 0$ (**runlength**=0), requiring just setting A[0]. (1 unit of effort.)
- There are $\sim \frac{n}{4}$ counter values with $A[0] = 1$ but $A[1] = 0$ (**runlength**=1), requiring flipping $A[0]$ to 0 and setting $A[1]$ to 1. (2 units of effort.)
- There are $\sim \frac{n}{8}$ counter values with $A[0 \ldots 1] = 1$ but $A[2] = 0$ (**runlength**=2), requiring flipping $A[0 \ldots 1]$ to 0 and setting $A[2]$ to 1. (3 units of effort.)
- There are $\sim \frac{n}{16}$ counter values with $A[0 \ldots 2] = 1$ but $A[3] = 0$ (**runlength**=3), requiring flipping $A[0 \ldots 2]$ to 0 and setting $A[3]$ to 1 (4 units of effort.), and so on...

Summing total time over n increments

- $T = \frac{n}{2} + \frac{2n}{4} + \frac{3n}{8} + \frac{4n}{16} \ldots \leq 2n$ (watch lecture for proof)

Thus, total time over n increments is bounded by $O(n)$, implying $O(1)$-time amortized cost for each **increment**!

Aggregate method of amortization on **binary counter**

Assume a sequence of n increments.

Counter value	A[7]	A[6]	A[5]	A[4]	A[3]	A[2]	A[1]	A[0]	Total cost
0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	1	1
2	0	0	0	0	0	0	1	0	3
3	0	0	0	0	0	0	1	1	4
4	0	0	0	0	0	1	0	0	7
5	0	0	0	0	0	1	0	1	8
6	0	0	0	0	0	1	1	0	10
7	0	0	0	0	0	1	1	1	11
8	0	0	0	0	1	0	0	0	15
9	0	0	0	0	1	0	0	1	16
10	0	0	0	0	1	0	1	0	18
11	0	0	0	0	1	0	1	1	19
12	0	0	0	0	1	1	0	0	22
13	0	0	0	0	1	1	0	1	23
14	0	0	0	0	1	1	1	0	25
15	0	0	0	0	1	1	1	1	26
16	0	0	0	1	0	0	0	0	31

- There are $\sim \frac{n}{2}$ counter values with $A[0] = 0$ (**runlength**=0), requiring just setting A[0]. (1 unit of effort.)
- There are $\sim \frac{n}{4}$ counter values with $A[0] = 1$ but $A[1] = 0$ (**runlength**=1), requiring flipping $A[0]$ to 0 and setting $A[1]$ to 1. (2 units of effort.)
- There are $\sim \frac{n}{8}$ counter values with $A[0 \ldots 1] = 1$ but $A[2] = 0$ (**runlength**=2), requiring flipping $A[0 \ldots 1]$ to 0 and setting $A[2]$ to 1. (3 units of effort.)
- There are $\sim \frac{n}{16}$ counter values with $A[0 \ldots 2] = 1$ but $A[3] = 0$ (**runlength**=3), requiring flipping $A[0 \ldots 2]$ to 0 and setting $A[3]$ to 1 (4 units of effort.), and so on...

Summing total time over n increments

- $T = \frac{n}{2} + \frac{2n}{4} + \frac{3n}{8} + \frac{4n}{16} \ldots \leq 2n$ (watch lecture for proof)

Thus, total time over n increments is bounded by $O(n)$, implying $O(1)$-time amortized cost for each **increment**!

Method 2: Accounting method of amortization

Accounting method (Amortized Analysis)

- Assume an imaginary 'coin-operated' data structure).
- Think of coins as buying **`credits`** to pay for some amount of work operating this data structure
- Each coin buys a **fixed amount of computation**.
- The **`true cost`** of each operation may vary.
- We can reason and assign some carefully chosen number of coins as part of each operation during this analysis – this is its **`amortized cost`**.
- Any extra coins (credit) can be **saved** and used to pay for future work.

Goal:

Show that any sequence of operations can be performed without running out of credits.

Accounting method: Key idea

- If we never go into debt, then:
- Total **actual cost** $\leq$ total **amortized cost**
- This holds for **any sequence of operations**.
- Credits are only an **analysis tool**:
  - They are **not implemented** in the data structure.

Allocation of 'coins' for key tasks of the **increment** operation in the **binary counter** data structure

The **increment** operation

```
1 b = 0
2 while b < k and A[b] == 1:
3     A[b] = 0 /*unset*/
4     b = b+1
5 if b < k:
6     A[b] = 1 /*set*/
```

- The increment operation consists of two atomic tasks:
  1. Setting a bit to 1 (line 6).
     - ★ This takes $O(1)$ time.
     - ★ Line 6 is executed at most once per increment.
  2. Unsetting bits to 0 (line 3).
     - ★ Only previously set bits are unset (line 3, via the while-condition on line 2).
     - ★ Line 3 may execute multiple times per increment until the while-condition becomes false.
     - ★ Each iteration of the loop (lines 2-4) costs $O(1)$ time.

Allocation of 'coins' for key tasks of the **increment** operation in the **binary counter** data structure

The **increment** operation

```
1 b = 0
2 while b < k and A[b] == 1:
3     A[b] = 0 /*unset*/
4     b = b+1
5 if b < k:
6     A[b] = 1 /*set*/
```

- Assume 1 coin pays for $O(1)$ work in any part of the **increment** operation.
- We assign $\lambda > 0$ coins whenever a bit is set (line 6).
- Setting a bit costs 1 coin, leaving $\lambda - 1$ coins as credit stored on that bit for future work.
- Can this stored credit pay for unsetting the bit later (during future increments)?
  - Yes. Each time a bit is reset, it is due to a while-loop iteration (lines 3–4), which costs $O(1)$ and is paid for by 1 coin.
  - Since each set bit carries $\lambda - 1$ credits, choosing $\lambda \geq 2$ ensures we never run out of credit.
- Hence, a sensible choice is $\lambda = 2$.

Accounting method applied on the binary counter

```
1 b = 0
2 while b < k and A[b] == 1:
3     A[b] = 0  /* unset */
4     b = b + 1
5 if b < k:
6     A[b] = 1  /* set */
```

- Assign amortized cost $\lambda = 2$ coins to setting a bit (line 6).
  - 1 coin pays for the actual write $0 \to 1$.
  - 1 coin is stored as credit on that bit.
- Assign 1 coin per loop iteration (lines 2-4).
  - This pays for unsetting a bit $1 \to 0$ (line 3).

- Each increment sets at most one bit (line 6).
- Each unset operation is paid for by credit stored on previously set bits.
- Hence no operation goes into debt (credit is always nonnegative).
- Therefore, each increment costs at most 2 **amortized** coins $\Rightarrow O(1)$ **amortized time**.
- Over n increments, total **amortized cost** is $O(n)$.

Method 3: Potential method of amortization

Historical note:

- The physicist's view of amortization was proposed by Daniel Sleator who was a PhD supervisor of Bob Tarjan.
- Together with Tarjan, they pioneered amortized analyses of data structures and algorithms.
- Important The banker's accounting and physicist's potential methods of amortization are essentially **equivalent**. The potential view provides a way to deal with fractional costs (for time), whereas the banker's view historically dealt with integer credits/debits. But important to note that both yield the same bounds on the true cost.

Potential method: Introduction

- Let D_0 be the initial state of some hypothetical data structure.
- Over n operations let the state of this data structure change as: $D_0 \to D_1 \to D_2 \to \ldots D_n$.
- Let τ_i be the actual time it takes to execute the i-th operation (changing the data structure from $D_{i-1} \to D_i$.

Potential method: Introduction

- Define a **potential function** Φ that maps each state D_i of the data structure to a real number $\Phi(D_i)$, called its **potential**.
  - A unit of potential represents **stored work (or credit)** that can pay for a **constant amount of computation**, sufficient to cover constant-time steps in operations.
  - **Intuition:** $\Phi(D_i)$ measures how much **future work has already been prepaid and stored as credit** in the current state.
- The **amortized cost/time** of the i-th operation is defined as:

$$a_i = \tau_i + \Phi(D_i) - \Phi(D_{i-1}).$$

  - **Intuition:** amortized cost = actual cost + change in stored credit.
    - ★ If potential increases, we are **prepaying credit for future operations**.
    - ★ If potential decreases, we are **using previously stored credit**.
- Summing over a sequence of n operations gives:

$$\sum_{i=1}^{n} a_i = \sum_{i=1}^{n} \tau_i + \Phi(D_n) - \Phi(D_0)$$

(To be derived in the lecture)

  - This shows that amortized analysis accounts for both **actual work** and changes in **stored credit**.

Potential method: Introduction (...cont'd)

- From the previous slide:

$$\sum_{i=1}^{n} a_i = \sum_{i=1}^{n} \tau_i + \Phi(D_n) - \Phi(D_0)$$

- In words, the total **amortized** cost equals total **actual** cost plus the net change in potential between final and initial states.
- The potential function $\Phi(\cdot)$ can be chosen freely depending on the problem (see next slide).

Choosing a potential function Φ

- An astute choice of Φ can yields an informative (tighter) amortized bound.
- In most cases (for convenience), the potential function is chosen such that where $\Phi(D_0) = 0$ and $\Phi(D_i) \geq 0$ for all i.
- While that is the matter of convenience, the key requirement is that $\Phi(D_n) - \Phi(D_0) \geq 0$ so that amortized cost upper bounds actual cost.
- Different valid choices of Φ can give different amortized bounds; the goal is to choose one that yields the tightest bound.

A **Potential function** for the **binary counter** data structure

- In the binary counter example, the data structure is an array of bits $A[0 \ldots k-1]$ of size k.
- Let $A_i[0 \ldots k-1]$ denote the state of the binary counter after the ith increment operation ($\forall i \geq 0$)).
- One could define a viable potential function as follows:

$$\Phi(A_i) = \texttt{bitsum}(A_i) = \sum_{b=0}^{k-1} A_i[b]$$

- With the above defined potential function:
  - $\Phi(A_0) = 0$.
  - $\Phi(A_i) \geq 0$ for all $i \geq 0$.
  - The maximum potential is k (when in some state, all k bits are set).

Potential method applied on the binary counter

We are using the potential function for this data structure specifically as

$$\Phi(A_i) = \texttt{bitsum}(A_i) = \sum_{b=0}^{k-1} A_i[b]$$

We saw on slide 23...

- ... the **amortized time** of an operation **in general** as:

$$a_i = \tau_i + \Phi(D_i) - \Phi(D_{i-1}).$$

- Over a sequence of n operations, we have the total amortized time as:

$$\sum_{i=1}^{n} a_i = \sum_{i=1}^{n} \tau_i + \Phi(D_n) - \Phi(D_0)$$

- Suppose that the i-th increment results in **unsetting** $(1 \to 0)$ some l trailing bits before potentially setting $(0 \to 1)$ the next most significant bit.
- The **true cost** τ_i is bounded by $l + 1$.

- Thus, **specifically for the binary counter**,

$$\Phi(A_i) = \begin{cases} \Phi(A_{i-1}) - l + 1, & \Phi(A_{i-1}) < k \\ \Phi(A_{i-1}) - l, & \Phi(A_{i-1}) = k \end{cases}$$

(Refer conditional statement on Line 5 of the pseudocode on slide 10).

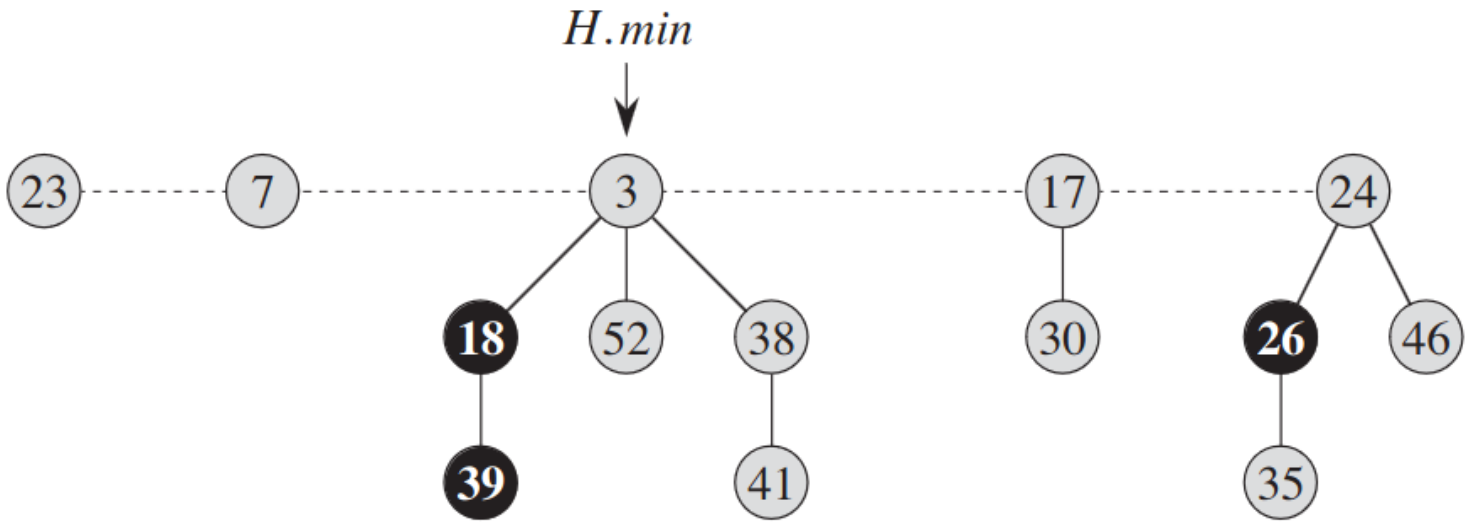
- The amortized cost of this operation under this potential is

$$\begin{aligned} a_i &= \tau_i + \Phi(A_i) - \Phi(A_{i-1}) \\ &\leq l + 1 + \Phi(A_{i-1}) - l + 1 - \Phi(A_{i-1}) \\ &\leq 2 = O(1). \end{aligned}$$

- Hence, the amortized bound on the true cost of a sequence of n **increment** operations is $O(n)$.

The performance of various **Fibonacci heap** operations using the **potential function** method

Defining a potential function for Fibonacci heap operations



- For a given Fibonacci heap H denote:
  - $\texttt{nTrees}(H)$ to mean the number of trees in the root list of H and
  - $\texttt{nMarked}(H)$ to mean the number of marked nodes in H.
- A **potential** $\Phi(H)$ of any Fibonacci heap H is chosen to be

$$\Phi(H) = \texttt{nTrees}(H) + 2(\texttt{nMarked}(H)).^*$$

- For example, the potential of the Fibonacci heap shown above is $\Phi(H) = 5 + 2(3) = 5 + 6 = 11$.

*(We will gain some intuition for this potential function in the upcoming slides.)

Amortized cost of Fibonacci heap's **insert** operation

Recall from slide 23...

- ... for a data strucuture D, **amortized time** is defined as:

$$a_i = \tau_i + \Phi(D_i) - \Phi(D_{i-1}).$$

- True time τ_i to perform any insert in the Fibonacci heap H is clearly $O(1)$.
- Inserting an element into H increases the number of trees (in the root level) by 1: $\texttt{nTrees}(H_i) = \texttt{nTrees}(H_{i-1}) + 1$.
- On the other hand, inserting an element into H does not change the number of marked nodes: $\texttt{nMarked}(H_i) = \texttt{nMarked}(H_{i-1})$.

- Hence, the potential difference going from $H_{i-1} \to H_i$ is:

$$\Delta\Phi = \Phi(H_i) - \Phi(H_{i-1}) = 1$$

- Thus the **amortized cost** of this operation under the defined potential (see slide 29) is:

$$\begin{aligned} a_i &= \tau_i + \Phi(H_i) - \Phi(H_{i-1}) \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

- This gives an amortised cost of $O(1)$ for each insert.
- From the above, since $\Phi(H_0) = 0$, it can be shown that the sum of amortized costs is $O(n)$ and bounds the true cost of any sequence of n inserts.

Amortized cost of Fibonacci heap's **extract-min**

- Recall that **extract-min** extracts the root node pointed by H.min, promotes its children (trees) into the root-list and then consolidates the heap such there is at most one tree of any given degree.
- Assume **extract-min** changes the state of the heap from H_{i-1} into H_i.
- The true cost of **extract-min** is bounded by $\text{degree}_{\text{max}}(H_{i-1}) + \texttt{nTrees}(H_{i-1})$.
- The **potential** before extracting node pointed by H_{i-1}.min is

$$\Phi(H_{i-1}) = \texttt{nTrees}(H_{i-1}) + 2(\texttt{nMarked}(H_{i-1})).$$

- The **potential** after extracting is

$$\begin{aligned}\Phi(H_i) &= \texttt{nTrees}(H_i) + 2(\texttt{nMarked}(H_i)) \\ &\leq \text{degree}_{\text{max}}(H_i) + 1 + 2(\texttt{nMarked}(H_{i-1})).\end{aligned}$$

The above inequality holds because (i) the nTrees(H_i) in the worst case after consolidation is $\text{degree}_{\text{max}}(H_i) + 1$, and (ii) no nodes become newly marked during extract-min and potentially the min node that has been extracted could have been previously marked, making nMarked$(H_i) \leq$ nMarked(H_{i-1}).

- Thus, the **amortized cost** of extract-min is:

$$\begin{aligned}a_i &= \tau_i + \Phi(H_i) - \Phi(H_{i-1}) \\ &\leq \underbrace{\text{degree}_{\text{max}}(H_{i-1}) + \texttt{nTrees}(H_{i-1})}_{\tau_i} + \underbrace{\text{degree}_{\text{max}}(H_i) + 1 - \texttt{nTrees}(H_{i-1})}_{\Phi(H_i)-\Phi(H_{i-1})} \\ &\leq \text{degree}_{\text{max}}(H_i) + \text{degree}_{\text{max}}(H_{i-1}) + 1\end{aligned}$$

- Since $\text{degree}_{\text{max}}(H)$ grows $O(\log H.\text{size})$, the amortized cost of **extract-min** is $O(\log(H.\text{size}))$ (H.size is the number of elements in the heap.).

Amortized cost of Fibonacci heap's **decrease-key**

- Let **decrease-key** operation on the state H_{i-1} of the heap results in $m \geq 0$ nodes being promoted to the root level.
- The true cost $\tau_i = m$ (as the effort is dominated by m nodes being promoted to the root).
- The **potential** before decrease-key was called is:

$$\Phi(H_{i-1}) = \texttt{nTrees}(H_{i-1}) + 2(\texttt{nMarked}(H_{i-1})).$$

- The **potential** after decreasing the key is

$$\begin{aligned} \Phi(H_i) &= \texttt{nTrees}(H_i) + 2(\texttt{nMarked}(H_i)) \\ &\leq \texttt{nTrees}(H_{i-1}) + m + 2(\texttt{nMarked}(H_{i-1}) - (m-1) + 1). \end{aligned}$$

  (inequality holds because at least $m-1$ and at most m previously marked nodes that were promoted to the root will now be unmarked, and the final cascading cut could have marked its previously unmarked parent)
- Thus, the **amortized cost** of decrease-key is:

$$\begin{aligned} a_i &= \tau_i + \Phi(H_i) - \Phi(H_{i-1}) \\ &\leq \underbrace{m}_{\tau_i} + \underbrace{\texttt{nTrees}(H_{i-1}) + m + 2(\texttt{nMarked}(H_{i-1})) - 2m + 4}_{\Phi(H_i)} - \underbrace{\texttt{nTrees}(H_{i-1}) - 2(\texttt{nMarked}(H_{i-1}))}_{\Phi(H_{i-1})} \\ &\leq 4 \end{aligned}$$

- Thus, this gives the amortized cost of $O(1)$ per **decrease-key** operation.

Intuition behind the chosen potential for Fibonacci – continued

- All Fibonacci heap operations potentially affect the number of trees in the root level and in the case of decrease-key, the number of marked nodes over all in the heap.
- So a viable potential function involves those two criteria.
- Yet, it does not explain how those constant multipliers in the specified potential function were chosen.
- Assume the potential of the Fibonacci heap H were more generally defined as:

$$\Phi(H) = \lambda_1(\texttt{nTrees}(H)) + \lambda_2(\texttt{nMarked}(H))$$

Intuition behind the chosen potential for Fibonacci Heap – continued

Revisiting the amortized analysis of decrease-key with this general form:

- The **potential** before decrease-key was called is:

$$\Phi(H_{i-1}) = \lambda_1(\texttt{nTrees}(H_{i-1})) + \lambda_2(\texttt{nMarked}(H_{i-1})).$$

- The **potential** after decreasing the key is

$$\begin{aligned}\Phi(H_i) &= \lambda_1(\texttt{nTrees}(H_i)) + \lambda_2(\texttt{nMarked}(H_i)) \\ &\leq \lambda_1(\texttt{nTrees}(H_{i-1}) + m) + \lambda_2(\texttt{nMarked}(H_{i-1}) - (m-1) + 1).\end{aligned}$$

Thus, we have the amortized cost bounding the true cost as:

$$\begin{aligned}a_i &= \tau_i + \Phi(H_i) - \Phi(H_{i-1}) \\ &\leq m + \lambda_1(\texttt{nTrees}(H_{i-1}) + m) + \lambda_2(\texttt{nMarked}(H_{i-1}) - (m-1) + 1) \\ &\quad -\lambda_1(\texttt{nTrees}(H_{i-1})) - \lambda_2(\texttt{nMarked}(H_{i-1})) \\ &\leq (1 + \lambda_1 - \lambda_2)m + 2\lambda_2\end{aligned}$$

This is not an informative amortized cost, as the r.h.s depends on m (the number of marked nodes that were promoted to the root level, which varies per operation). However, if one were to choose values for λ_1 and λ_2 more astutely such that $1 + \lambda_1 - \lambda_2 = 0$, one could make that term involving m disappear, giving a $O(1)$ cost. Thus, $\lambda_1 = 1$ and $\lambda_2 = 2$ becomes a natural choice.

Next week: B-trees